Strawberry vs Vanilla

5 instances · Download all inputs (.zip)

Strawberry vs Vanilla

You are given a rectangular grid of cells. Each cell must be filled with exactly one of two flavors: **strawberry** or **vanilla**. In the output, we will represent strawberry by `1` and vanilla by `0`.

Cells of the same flavor that touch orthogonally (up, down, left, right) form a **block** of that flavor. Your task is to assign a flavor to every cell so that strawberry blocks are very regular, while vanilla blocks are deliberately irregular.

The constraints are:

- Every **strawberry block** must form a perfect axis-aligned rectangle (including squares). That is, if you look only at strawberry cells, each connected component must be a solid `h x w` rectangle, with no holes and no missing corners.
- Every **vanilla block** must *not* be a rectangle. Any connected component consisting only of vanilla cells must have a non-rectangular shape. It can bend, have "bites" missing, or snake around the grid – but it must not coincide with any full `h x w` rectangle.
- Some cells contain **clues** in the input. A clue is a positive integer written in a cell. Each clue specifies the size of the block (strawberry or vanilla) that contains that cell.
  - The number counts *all* cells in that block (including any other clue cells in the same block).
  - A block may contain several clues; in that case all those clues must show the same number, equal to the block size.
  - A block may also contain no clues at all.

Goal

Determine which cells are strawberry and which are vanilla, satisfying all the above conditions.

Input Format

The input describes one grid.

- The first line contains two integers, `R` and `C`: the number of rows and the number of columns of the grid.
- Then follow `R` lines, each containing `C` integers.

For row `r` and column `c`, the given integer is: - `0` if there is no clue in that cell. - A positive integer `v` if there is a clue stating that the block containing this cell (strawberry or vanilla) has exactly `v` cells.

Rows are numbered from top to bottom, `1..R`. Columns are numbered from left to right, `1..C`.

The input will always describe a valid puzzle instance; in particular, there is at least one solution.

Output Format

Output `R` lines, each with `C` integers separated by single spaces.

- Print `1` in position `(r,c)` if that cell is strawberry.
- Print `0` in position `(r,c)` if that cell is vanilla.

The resulting grid must satisfy all rules:

- Every connected block of `1` cells is a rectangle.
- No connected block of `0` cells is a rectangle.
- For every clue in the input, the number equals the total number of cells in the block (of either flavor) that contains that cell.

Any output that violates these conditions will be judged incorrect.

Example

Instance in input:

```
7 7
0 6 0 0 0 0 0
0 0 0 0 0 0 3
0 0 0 3 0 0 0
0 0 6 1 5 0 0
0 0 0 4 0 0 0
5 0 0 0 0 0 0
0 0 0 0 0 4 0
```

Solution in output:

```
1 0 0 1 1 0 0
1 0 1 0 0 1 0
1 0 1 0 1 0 1
1 0 0 1 0 0 1
0 1 1 0 1 0 0
0 1 1 0 0 1 1
0 0 0 1 0 1 1
```

Visualizer

A graphical visualization can be obtained on ASP Chef.

Instances

Submit all instances at once

Upload one .zip containing a file per instance, each named after its instance number (e.g. `instance.1.out`, `2.txt`).

Browse... No file selected.

Submit all

Your code

Upload a .zip of the code your team used to generate the answers for this problem. Re-

uploading replaces the previous file.

Browse... No file selected.

Upload code

Instance 1

▸ Show input

Paste your answer

Paste output here, or upload a file below

or upload a file

Browse... No file selected.

Submit instance 1

Instance 2

▸ Show input

Paste your answer

Paste output here, or upload a file below

or upload a file

Browse... No file selected.

Submit instance 2

Instance 3

▸ Show input

Paste your answer

Paste output here, or upload a file below

or upload a file

Browse... No file selected.

Submit instance 3

Instance 4

▸ Show input

Paste your answer

Paste output here, or upload a file below

or upload a file

Browse... No file selected.

Submit instance 4

Instance 5

▸ Show input

Paste your answer

Paste output here, or upload a file below

or upload a file

Browse... No file selected.

Submit instance 5